For calculating the greatest common divisor, Euclid's algorithm required several modulo operations. On hardware lacking a dedicated divider, one modulo instruction costs tens to hundreds of clock cycles. In contrast, the Binary GCD algorithm relied on bitwise shifts and subtractions. These are basic ALU operations taking just one clock cycle each. Therefore, despite requiring more steps, Binary GCD consumes vastly fewer total CPU cycles, making it significantly faster on division-constrained hardware.